Probability Midterm 1

50 minutes • 100 points • Six problems

Name: _____ **Date:** _____

Instructions. Answer all six problems and show your reasoning. Exact answers are acceptable; binomial coefficients need not be evaluated. No calculator is needed. Use the spaces provided; clearly label any work continued on the back. In Problem 6, use the guided steps rather than quoting a coupon-collector formula. Reserve time to check your work.

1. Sets — 10 points

Let $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, and $B = \{1, 2, 3, 4\}$. Take all complements relative to U.

(a) Find $A \cap B$, $A \setminus B$, and $(A \cup B)^c$. (6 points)

(b) For these sets A, B, is the following statement true for every $C \subseteq U$? Justify your answer with a proof or a counterexample. (4 points)

$$C \subseteq A \cup B \quad \Longrightarrow \quad C \subseteq A \text{ or } C \subseteq B.$$

2. A probability space — 10 points

Let $\Omega = \{a, b, c, d\}$, with every subset of Ω an event. The probabilities of the individual outcomes a, b, c, d are $k, 2k, 3k, 4k$, respectively.

(a) Find k, explaining the probability axiom you use. (4 points)

(b) Let $E = \{a, c\}$ and $F = \{b, c\}$. Find $\mathbb{P}(E)$, $\mathbb{P}(E \cap F)$, and $\mathbb{P}(E^c)$. (6 points)

3. Inclusion–exclusion **15 points**

In a class of 40 students, 18 study French, 16 Spanish, and 12 German. Seven study both French and Spanish, five both French and German, and four both Spanish and German. Two study all three languages. **The pairwise counts include students who study all three.**

(a) How many students study at least one of these languages? Show your inclusion–exclusion calculation. (11 points)

(b) How many students study none of these languages? (4 points)

4. Balls in an urn **20 points**

An urn contains five red, four blue, and three green balls. Four balls are selected uniformly without replacement: every four-ball subset of the twelve individual balls is equally likely.

(a) Find the probability of selecting exactly two red, one blue, and one green ball. (10 points)

(b) Find the probability of selecting at least one green ball. (10 points)

5. A density, its mean, and its variance

25 points

A continuous random variable X has probability density

$$f(x) = \begin{cases} cx^3, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant. Show the integrals you use.

(a) Find c. (5 points)

(b) Compute $\mathbb{E}[X]$. (8 points)

(c) Compute $\mathbb{E}[X^2]$, and hence $\mathrm{Var}(X)$. (12 points)

6. Collecting four toy types **20 points**

Each cereal box contains one toy. There are four toy types, each equally likely, and different boxes have independent contents. Duplicates are possible. Starting with no toys, let T be the number of boxes opened up to and including the first box that completes the collection of all four types.

(a) After exactly k distinct types have been collected, let p_k be the probability that the next box contains a new type. Find p_0, p_1, p_2, p_3, and explain your reasoning. (6 points)

(b) Let W_k be the number of additional boxes needed to go from exactly k collected types to $k+1$ collected types, including the box containing the new type. Find $\mathbb{E}[W_k]$ for $k = 0, 1, 2, 3$. (8 points)

You may use: A geometric waiting time that counts trials through the first success, with success probability p, has mean $1/p$.

(c) Express T in terms of W_0, W_1, W_2, W_3 and compute $\mathbb{E}[T]$. State the expectation rule you use. (6 points)

End of paper. Check that you have answered all six problems.